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Local cation-clamping distorts and softens RNA duplex

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Multivalent cations are essential for DNA and RNA structures and their functions. However, how multivalent cations and their accompanying anions jointly reshape nucleic-acid mechanics, and why DNA and RNA respond so differently, remain unclear. Here we show using single-molecule magnetic tweezers that increasing multivalent cation concentrations first softens and then stiffens DNA due to charge inversion, whereas RNA first stiffens and then softens by over twofold. Our all-atom simulations reproduce these effects and reveal a physical mechanism: at low concentrations, multivalent cations clamp the RNA major groove, stiffening RNA; at higher concentrations, anions disrupt this groove clamping and promote local cation-clamping that distorts and softens RNA. The mechanism also explains changes in stretching modulus, contour length, and twist-stretch coupling. Our findings reveal that local cation-clamping sharply softens RNA and highlight the significant roles of anions in modulating RNA structure, providing a framework for tuning nucleic-acid mechanics in vitro for RNA-related applications.

The structural regulation and biophysical properties of DNA and RNA are critical for various biological processes such as replication, transcription, and translation[1–5]. As DNA and RNA are highly negatively charged, cations critically affect their biophysical properties including net charges, structures, and elasticities[6–8]. Monovalent cations such as Na^+ and K^+ neutralize the negative charges of DNA and RNA duplexes, thus increases their stabilities and decreases their bending stiffness[9–14]. Divalent cations such as Mg^{2+} and Ca^{2+} stabilize DNA duplexes by neutralizing their negative charges[15,16]. High-valent cations such as $CoHex^{3+}$ and $Spermidine^{3+}$ (SPD^{3+}) soften DNA and induce DNA condensation by neutralizing the negative charges of DNA and attracting different DNA segments[17–22]. At high concentrations, excessive binding of high-valent cations can reverse the charges of DNA, which leads to DNA re-stiffening[19,23,24] and the reentrainment of DNA condensates[18,25,26]. Compared to DNA, RNA typically exhibits greater bending rigidity and lower stretch rigidity. Furthermore, recent magnetic tweezer (MT) experiments and molecular dynamics (MD) simulations have revealed that RNA rigidity in bending and stretching dimensions can be significantly enhanced by multivalent ions, primarily due to the formation of cation-clamping across the RNA major groove by multivalent ions[23,24]. In addition, recent MD simulations integrated with solution wide angle x-ray scattering experiments demonstrated that an RNA structure shows marked sensitivity to the valence and type of its bound cations[7].

Multivalent salts at high concentration bring more cations and anions into the vicinity of nucleic acid, resulting in more complex ion-nucleic acid interactions and ion-ion correlations[27–31]. Recently, the specificity and strength of anion-nucleic acid interactions have been explored as well as the factors modulating them, such as anion type, nucleic acid sequence and structure, and other solution conditions[28,30,32]. There is still a lack of a deep understanding of how anions along with cations affect the structure and elasticity of RNA duplex.

Results

Elasticities of RNA and DNA from experiments

Using MT experiments, we measured the effects of multivalent salts, including $SPDCl_3$, $CaCl_2$, $MgCl_2$, and $SrCl_2$, on the elasticities of RNA and DNA duplexes through force-extension (*F-x*) curves (Fig. 1C–F). We fitted each *F-x* curve to an approximation formula[33] for the extensible worm-like-chain model following previous works[12,34–39]:

$$x = NL_c[1 - (k_BT/FP)^{1/2}/2 + F/K]. \quad (1)$$

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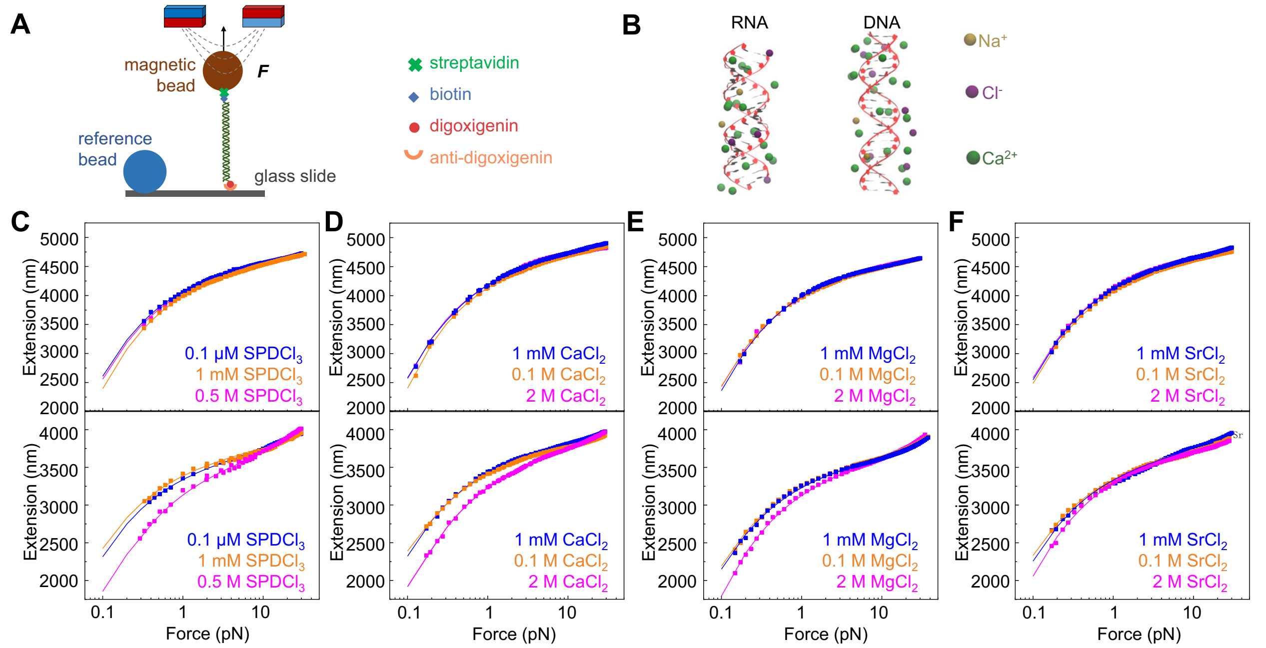

Fig. 1 | MT experiments and MD simulations to determine elasticities of RNA and DNA. A The MT setup contains a pair of NdFeB magnets to stretch an RNA or DNA molecule anchored between a glass slide and a paramagnetic microbead. **B** The initial structures of RNA and DNA in MD simulations. **C–F** Representative *F-x* curves of RNA and DNA at various $SPDCl_3$, $CaCl_2$, $MgCl_2$ or $SrCl_2$ concentrations. The data in each panel are from the same molecule.

Here, the two measured variables x is the extension of the molecule and F is the force. The fitting parameters P is the bending persistence length, L_c is the contour length per base pair and K is the stretching modulus, $N = 13751$ is the number of base pairs. Each individual curve fit by the WLC varying all three parameters simultaneously and then these values were averaged for each condition along with standard error based on the absolute best fit. At each salt condition, we considered the effect of the refractive index of the buffer solution following our previous works to obtain the extension of RNA and DNA[40–43] (Supplementary Fig. 1).

Also, we measured dL/dN at 8 pN to characterize twist-stretch coupling using torsion-constrained molecules following previous works[40,44,45] (Supplementary Fig. 2). Here, dN is the change in helical turn and dL is the corresponding change in DNA or RNA extension. The details of the preparation of the RNA and DNA constructs for MT experiments can be found in our previous works[24,39,40,42,43], please see Supplementary Fig. 3 and Supplementary Method 1.

As shown in Fig. 2, the experimental P_{DNA} initially decreases with increasing salt concentration until reaching a critical concentration, after which it increases, likely due to charge inversion. As we reported previously[17], DNA condensation may occur in the presence of $SPDCl_3$ at forces lower than a nucleation force (less than 0.3 pN) depending on the concentration of $SPDCl_3$. To avoid this influence on force-extension curves, we fitted the force-extension curves at higher forces without DNA condensation. The critical concentration of $SPDCl_3$ agrees with the concentration where the net charge of DNA is fully neutralized[18,19,46]. The values of K, L_c, and dL/dN of DNA remain largely unchanged across different salt concentrations. (Fig. 2A–D).

Opposite to P_{DNA}, the experimental P_{RNA} demonstrated an initial increase and then a sharp decrease with increasing salt concentration. When $SPDCl_3$ increased from 1 μM to 1 mM and then to 0.5 M, P_{RNA} increased from 64.4 ± 2.2 nm to 72.6 ± 1.6 nm and then sharply decreased to 32.2 ± 0.8 nm. Similarly, when $CaCl_2$ increased from 0.1 mM to 10 mM and then to 2 M, P_{RNA} increased from 53.5 ± 1.4 nm to 57.7 ± 1.4 nm and then dramatically decreased to 30.3 ± 1.3 nm. Simultaneously, L_c of RNA duplex decreased and then increased, and K increased and then decreased along with increasing the salt concentration. Additionally, dL/dN of RNA reverses to positive and then back to negative again as salt concentration increases. The increase in P and K, the decrease in L_c, and the first reversal of dL/dN for RNA at the low salt concentrations can be explained by the cation-clamping of the RNA major groove[23,24,40,47]. However, the counterintuitive results of P, K, L_c and dL/dN for RNA duplex at high salt concentrations are out of our anticipation and certainly require a mechanism.

Elasticities of RNA and DNA from simulations

To investigate the elasticity of DNA and RNA at high salt concentrations, we performed 600-ns MD simulations; please see Supplementary Figs. 4-5, Supplementary Method 3 and our previous works[43] for more details and the calculations of elastic parameters of RNA and DNA. As shown in Fig. 2A, our simulations reproduced the changes in the experimental P_{RNA} and P_{DNA}. Correspondingly, as salt concentration increased, P_{DNA} from simulations initially decreased and then increased, while P_{RNA} showed the opposite trend: it increased first and then sharply decreased. Specifically, as $SPDCl_3$ increased from 1 μM to 1 mM and then to 500 mM, P_{RNA} increased from 60.5 ± 4.1 to 69.9 ± 3.2 nm and then sharply decreased to 41.2 ± 3.3 nm. Similarly, with $CaCl_2$ increasing from 100 μM to 100 mM and then to 2 M, P_{RNA} increased from 55.5 ± 3.8 to 58.4 ± 1.6 nm, then dramatically dropped to 36.6 ± 1.4 nm. Also consistent with experiments, L_c of RNA duplex first decreased then increased, and K first increased and then decreased as salt concentration increased (Fig. 2F–G). The simulations also captured the two reversals in dL/dN of RNA duplex (Fig. 2H). For $MgCl_2$ and $SrCl_2$, our experiments and simulations obtained the same trends of the salt dependence of DNA and RNA elasticities (Fig. 2I–P). It should be noted that L_c of RNA duplex from MD simulations is shorter compared to experiment while the trend of changes with ion concentration remains consistent.

Binding of multivalent cations to DNA and RNA

We then analyzed the ion-binding to DNA and RNA (Fig. 3, Table 1, and Supplementary Fig. 6). Figure 3A shows that P_{DNA} closely correlates with the net charge of DNA together with binding ions. The P_{DNA} decreases to the lowest values at critical concentrations (1 mM for $SPDCl_3$ and 100 mM for $CaCl_2$), where the charge of DNA is nearly neutralized (Fig. 3A). At higher salt concentrations of $SPDCl_3$ and $CaCl_2$, the net charge of DNA reversed, consistent with re-stiffening of DNA. For a comparison, the net

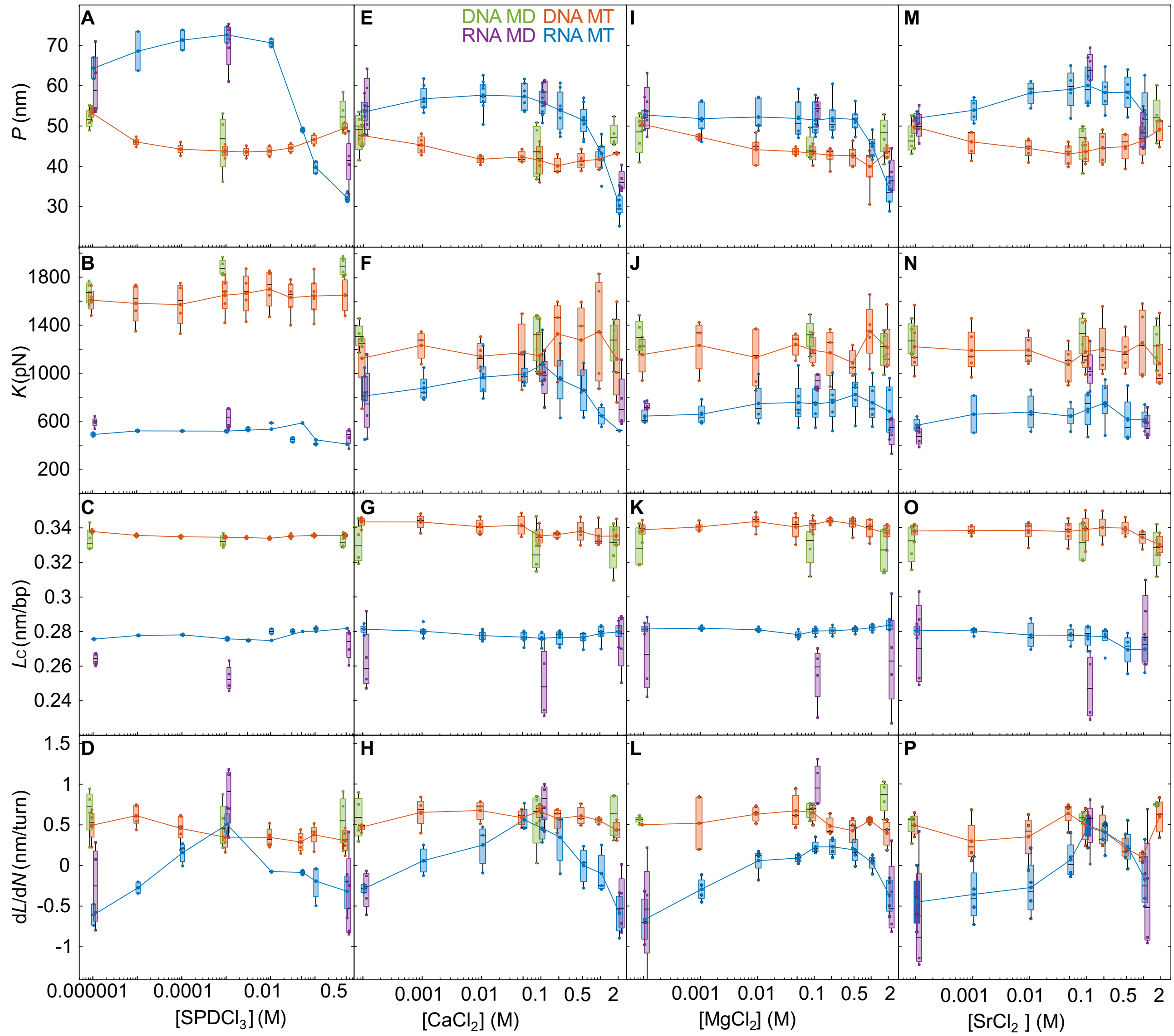

Fig. 2 | Effects of multivalent salts on the elasticities of DNA and RNA. Persistence length P, stretch modulus K, contour length L_c, and dL/dN of RNA and DNA as functions of $SPDCl_3$ (**A–D**), $CaCl_2$ (**E–H**), $MgCl_2$ (**I–L**), and $SrCl_2$ (**M–P**) concentrations. Each box represents the data distribution from at least three molecules in MT experiments or from four time-intervals in MD trajectories. For MT experimental data, the mean values are connected by solid lines.

charge of RNA did not obviously reverse at high $SPDCl_3$ or $CaCl_2$ concentrations (Fig. 3A), which is consistent with that P_{RNA} did not increase but sharply decreased (Fig. 2A).

To explore the counterintuitive dependence of RNA elasticity on multivalent salt concentrations, we analyzed the spatial standard deviation of major groove width along the axis of RNA, which is defined as σ_M (Fig. 3 and Supplementary Figs. 7–10). The σ_M reflects irregularity of the major groove due to cation-clamping of RNA (Fig. 3B-C)[23,24,43,48]. At low concentrations (≤ 1 mM $SPDCl_3$ or ≤100 mM $CaCl_2$), multivalent cations bind to the major groove of RNA (Fig. 3B, C), creating cation-clamping that significantly reduces σ_M and decreases the bending angle (Fig. 3D, E and Supplementary Fig. 9), thereby increasing P_{RNA}. To better illustrate the binding of ions and the structural changes of RNA duplexes, we also present the ion distributions around RNA at 300 ns, 400 ns, and 500 ns and the specific values of the major groove over MD time (Supplementary Fig. 10-11). Also, both experiments and simulations confirmed the twist-stretch coupling reversal at 1 mM SPD^{3+} or 100 mM Ca^{2+} (Fig. 2D, H), consistent with the cation-clamping mechanism.

Furthermore, we examined the distribution of cation-clamping along the axis of RNA duplex at different salt concentrations (Fig. 3B–D and Supplementary Fig. 11). At 0.1 mM $CaCl_2$, the major groove of RNA kept wide and stable over MD time, which indicates very few cation clamps formed. At 1 μM and 1 mM $SPDCl_3$ or 100 mM $CaCl_2$, the major groove width became narrow and remained stable over MD time, which means uniform cation clamps formed, consistent with the increase in P_{RNA} (Fig. 2A and Table 1). The groove widths at some indices are small but non-zero, as shown in Supplementary Fig. 11. However, at high concentrations (0.5 M $SPDCl_3$ or 2 M $CaCl_2$), σ_M increased, and the bending angle became larger (Fig. 3E), which is consistent with the sharp decrease in P_{RNA} (Fig. 2A). The strong correlations between the distribution of cation clamps, σ_M, and bending angle of RNA indicate that P_{RNA} is strongly related to the formation and destruction of cation clamps of the major groove (Table 1).

To further understand the role of cation and anion in the formation and disruption of cation-clamp, we calculated the ion distributions around the RNA and DNA duplexes, and the binding ions were partitioned into those binding around phosphate groups, and those binding in major and minor grooves, in analogy to refs. 23 and [51]. Specifically, the number of binding ions into the major and minor grooves were calculated as those within a helical distance of 10 Å and in the respective major and minor grooves, and the number of binding ions around phosphate groups were

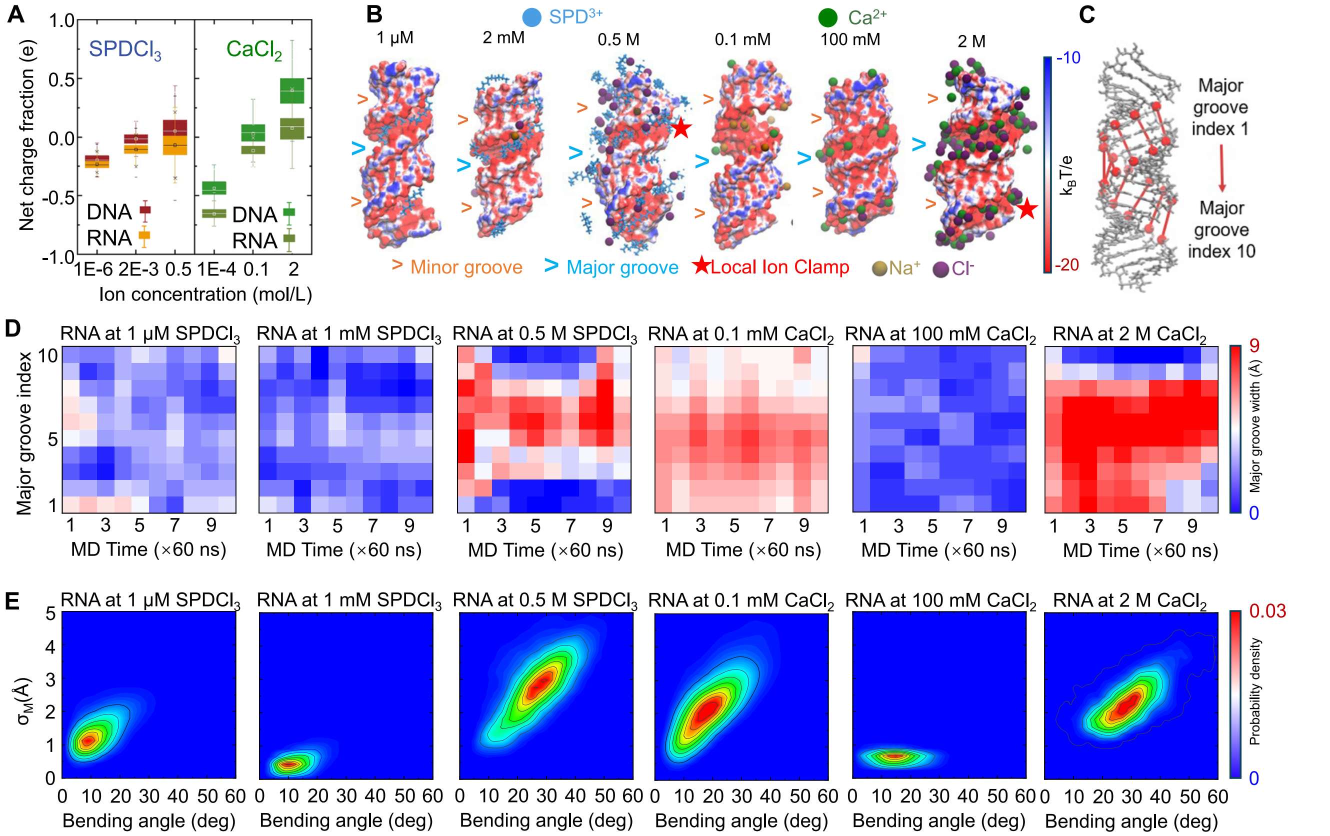

Fig. 3 | Effects of cation-clamping on RNA structure from simulations. A Net charges of RNA and DNA as functions of SPDCl$_3$ and CaCl$_2$ concentrations. Each box represents the data distribution from all frames in the last 400-ns MD trajectories. **B** Representative RNA structures showing cation distributions from MD simulations and surface electrostatic potentials from the PB solver of APBS. The structures are obtained from the final MD frames. **C** The major groove index used in the calculation. **D** Major groove width versus MD time at typical SPDCl$_3$ and CaCl$_2$ concentrations. The major groove widths were calculated by Curves +, which is widely used for the structural analysis of both DNA and RNA duplexes. **E** Relationships between the spatial standard deviation (σ_M) of major groove width along the axis of RNA (see panel **C**), and bending angle at typical SPDCl$_3$ and CaCl$_2$ concentrations.

Table 1 | Distributions of Ca^{2+}, SPD^{3+} and Cl^- on DNA and RNA duplexes and the averages and standard deviations of the helical rise, helical twist, and the width of the grooves revealed by MD simulations

Charge fraction		Binding Ca^{2+} or SPD^{3+}			Binding Cl^-			Helical rise (Å)	Helical twist (deg)	Major groove width (Å)
DNA		Phosphates	Major groove	Minor groove	Phosphates	Major groove	Minor groove			
CaCl$_2$	0.1 mM	0.30 ± 0.04	0.04 ± 0.03	0.02 ± 0.01	\	\	\	3.31 ± 0.06	35.02 ± 0.97	11.75 ± 0.67
	100 mM	0.87 ± 0.11	0.16 ± 0.08	0.06 ± 0.02	−0.03 ± 0.03	-0.03 ± 0.03	0.00 ± 0.01	3.29 ± 0.08	33.40 ± 0.87	11.89 ± 0.79
	2 M	2.91 ± 0.13	0.69 ± 0.13	0.17 ± 0.04	−1.72 ± 0.13	-0.58 ± 0.10	-0.08 ± 0.01	3.28 ± 0.08	33.17 ± 0.85	12.69 ± 0.75
SPDCl$_3$	1 μM	0.40 ± 0.06	0.15 ± 0.06	0.19 ± 0.02	\	\	\	3.29 ± 0.06	34.01 ± 0.95	12.26 ± 0.70
	2 mM	0.51 ± 0.07	0.17 ± 0.08	0.19 ± 0.02	−0.01 ± 0.02	-0.01 ± 0.01	0.00 ± 0.01	3.30 ± 0.06	34.54 ± 0.96	12.08 ± 0.75
	0.5 M	1.20 ± 0.12	1.07 ± 0.10	0.04 ± 0.01	−0.75 ± 0.13	−0.51 ± 0.10	0.00 ± 0.01	3.29 ± 0.06	34.55 ± 0.95	12.14 ± 0.72
RNA		Phosphates	Major groove	Minor groove	Phosphates	Major groove	Minor groove			
CaCl$_2$	0.1 mM	0.07 ± 0.04	0.10 ± 0.05	0.020.03	\	\	\	2.76 ± 0.11	32.61 ± 0.58	5.50 ± 1.28
	100 mM	0.36 ± 0.06	0.50 ± 0.07	0.11 ± 0.07	−0.08 ± 0.02	0.00 ± 0.01	0.00 ± 0.01	2.32 ± 0.08	34.82 ± 0.78	1.72 ± 0.53
	2 M	1.72 ± 0.13	0.38 ± 0.10	0.17 ± 0.10	−0.97 ± 0.09	−0.15 ± 0.05	−0.08 ± 0.03	2.74 ± 0.09	31.45 ± 0.76	7.09 ± 1.02
SPDCl$_3$	1 μM	0.23 ± 0.03	0.32 ± 0.04	0.16 ± 0.03	\	\	\	2.42 ± 0.09	33.51 ± 0.62	3.31 ± 1.05
	2 mM	0.32 ± 0.04	0.41 ± 0.04	0.18 ± 0.03	-0.01 ± 0.01	−0.01 ± 0.01	0.00 ± 0.01	2.37 ± 0.10	32.85 ± 0.59	2.07 ± 0.70
	0.5 M	0.91 ± 0.10	0.46 ± 0.05	0.31 ± 0.06	-0.62 ± 0.12	−0.11 ± 0.05	−0.03 ± 0.03	2.52 ± 0.11	33.19 ± 0.71	5.31 ± 1.47

calculated as those other than the ions in grooves and within a radial distance of 6 Å from phosphate atoms. At low salt concentrations (e.g., 0.1 mM CaCl$_2$), cations predominantly bind to the phosphate groups for DNA while prefer to bind into the major groove for RNA, with almost no anion binding observed. When salts increased to a higher concentration (e.g., 100 mM CaCl$_2$), the binding cations to the phosphate groups of DNA and into the major groove of RNA are both enhanced, corresponding to the enhanced ion neutralization for DNA and enhanced cation-clamping across the major groove of RNA. These cause a slight softening of DNA and an increase in the bending stiffness of RNA duplex. However, at even high salt concentrations

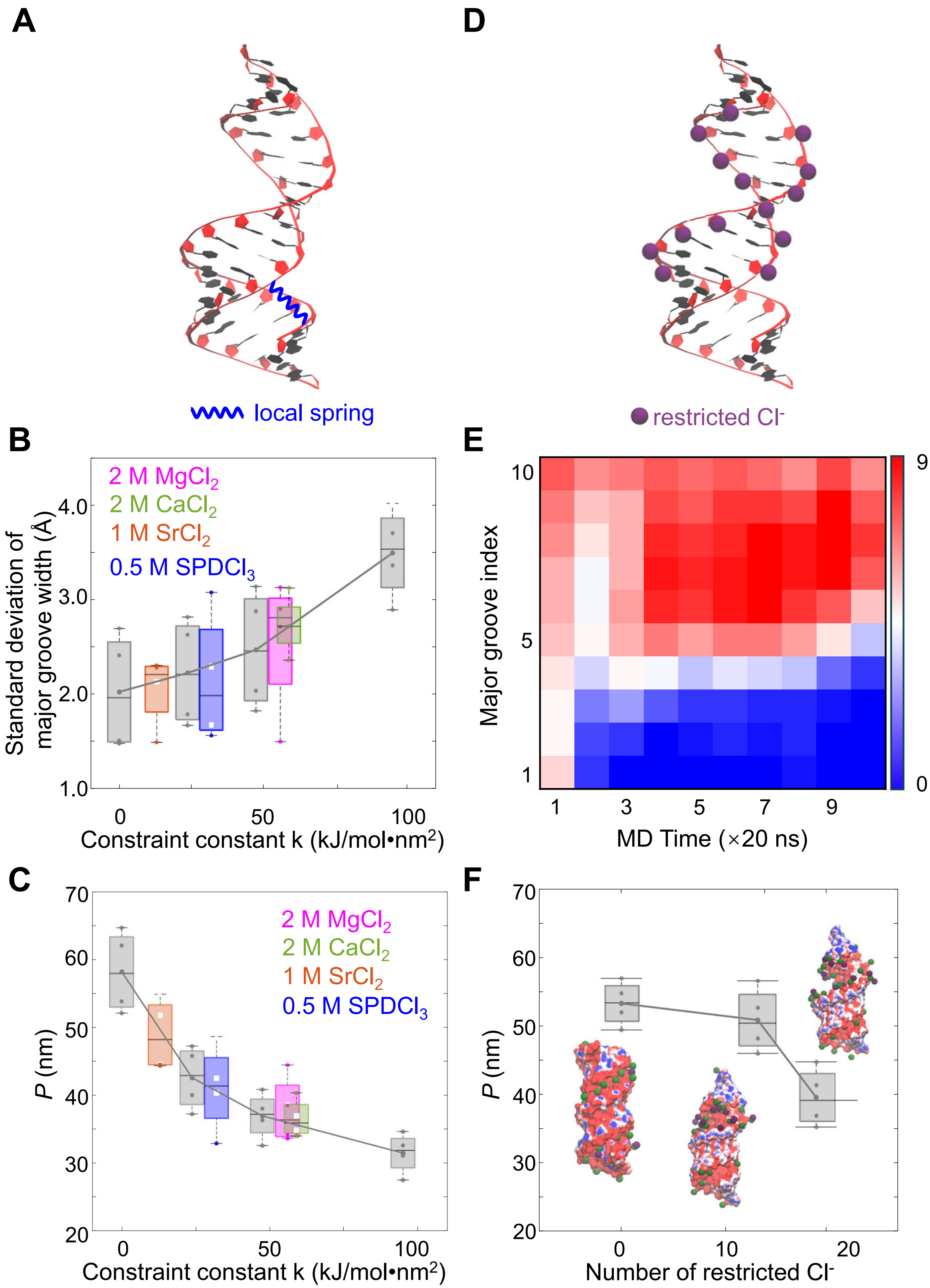

Fig. 4 | Effects of local cation-clamping and anion invasion on P_{RNA}. **A** The artificial MD simulation systems with a local spring clamping the major groove of RNA to simulate local cation-clamping. **B** Relationship between σ_M and the spring constant *k*. **C** Relationship between P_{RNA} and *k*, demonstrating the local-clamping-induced RNA softening. **D** The artificial MD simulations with restricted Cl^- ions around phosphate groups at 100 mM $CaCl_2$ to simulate anion invasion. **E** Time-evolution of major groove width of RNA from the MD simulation with restricted 20 Cl^- ions around phosphate groups. **F** Relationship between P_{RNA} and restricted Cl^- number for the synthetic MD systems, demonstrating strong consistency between the results from experiments. In **B**, **C** and **F** data estimated for local cation-clamps formed by different multivalent salts based on the σ_M from the MD simulations are shown in colored symbols, and data from artificial "spring" are shown in gray symbols. Each box represents the data distribution from four time-intervals in MD trajectories.

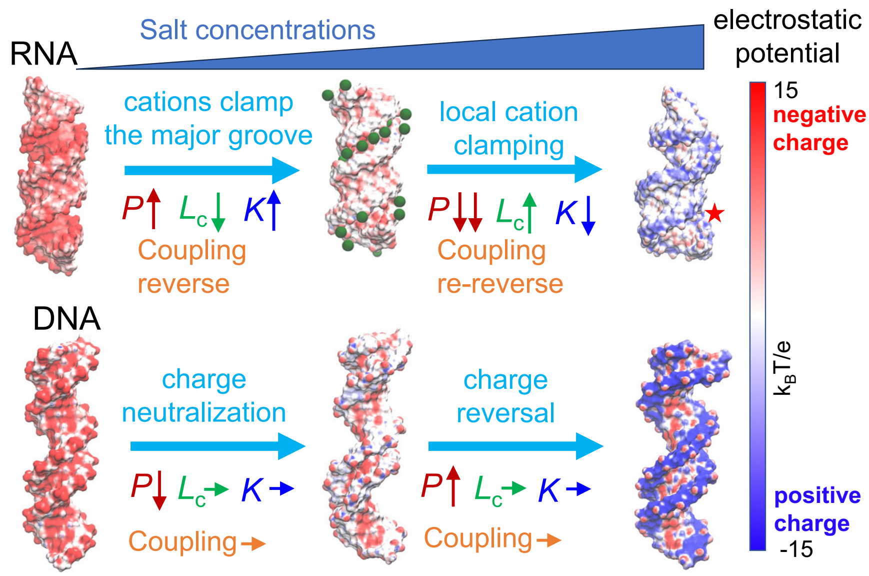

Fig. 5 | Mechanism by which multivalent salts influence DNA and RNA elastic parameters across wide salt concentration ranges. The changes in four elastic parameters, persistence length P, contour length L_c, stretching modulus K and twist-stretch coupling were shown. The arrows "↑" indicate an increase, "↓" a decrease, "↓↓" a sharp decrease, and "→" no change. Representative structures and surface electrostatic potentials of RNA and DNA duplexes were calculated at 0.1 mM, 100 mM, and 2 M $CaCl_2$ using the PB solver in APBS.

(e.g., 2 M $CaCl_2$), a significant number of invasion anions are observed around both DNA and RNA, especially near the phosphate groups and the major groove; see Table 1. Such strong anion invasion in the vicinity of RNA would disrupt the well-organized cation-clamp in the major groove of RNA (Figs. 3–4), causing a sharp decrease in RNA bending stiffness and a subsequent transition in twist-stretch coupling; please see Fig. 2.

To test the effects of local cation-clamping on the bending stiffness of RNA, we introduced an artificial "spring" between two phosphate atoms across the major groove of RNA in simulations (Fig. 4A). We found increasing the spring constant k elevated σ_M (Fig. 4B) and sharply reduced P_{RNA} (Fig. 4C). Additionally, we calculated the effective k of different multivalent cations based on σ_M from the simulations, and plotted the relationship between P_{RNA} and effective k for various multivalent cations at high concentrations (Fig. 4C). The simulation results showed a strong correlation between k and P_{RNA} both for the systems with multivalent cation but without "spring" (Fig. 2A) and for those from our artificial spring model (Fig. 4B, C). The stronger the local cation-clamping, the greater its effect on P_{RNA}. Additionally, we also selected another phosphate atom pair for additional MD simulations and obtained the similar results (Supplementary Fig. 12).

Anions at high concentrations invade RNA and lead to local cation-clamping

Simulations showed that, at high concentrations, anions can invade RNA duplex (Fig. 3B and Table 1), destroying most cation clamps and resulting in an irregular local cation-clamping distribution (Fig. 3D and Supplementary Fig. 13). This result is supported by twist-stretch coupling, which re-reverses at 0.5 M $SPDCl_3$ or 2 M $CaCl_2$ (Fig. 2D, H), suggesting that most locations of RNA major groove return to unclamped and is supported by the distribution of anions around dsRNA (Supplementary Fig. 13). To test the origin of local cation-clamping, we designed a synthetic system with 14 or 20 restricted Cl^- ions around phosphate groups of the 20-bp RNA duplex to simulate anion invasion in simulations (Fig. 4D), and found that the distribution of the RNA major groove became irregular, indicating a local cation-clamping effect (Fig. 4E). At the locations with Cl^-, the major groove became wide, indicating the loss of cation clamps, while major groove remained narrow at locations without Cl^-. As the Cl^- number increased, P_{RNA} decreased (Fig. 4F), with the structure deformation closely similar with that at high multivalent salt concentrations (Fig. 3B). The artificial MD simulations basically reproduce effect of the anion accumulation around phosphate groups of RNA duplex at high concentrations. However, anions were forcibly restrained around certain phosphate groups in the artificial simulations at relatively lower salt concentration since very high salt naturally causes anion invasion and sharp softening of RNA. Here, we did not make a direct comparison between the unrestrained MD ones at high salts and the artificial MD simulations at relatively lower salt since they involved different ion backgrounds. The restriction of Cl^- around RNA destroys cation-clamps in this synthetic system supports our above analyses that anion invasion leads to local cation-clamps that sharply reduce P_{RNA}.

Discussion

Using experiments and simulations, we investigated how multivalent salts influence four elastic parameters of DNA and RNA duplexes across wide salt concentrations (Fig. 5). Multivalent cations preferentially bind to the phosphate groups of DNA, initially neutralizing charges and decreasing P_{DNA} before charge inversion. In contrast, multivalent cations bind preferentially to the major groove of RNA, forming cation clamps that dominate P_{RNA} modulation (Supplementary Table 1). At low concentrations, cation-clamps stiffen RNA. As cation concentrations further increase, anions invade RNA and disrupt these cation clamps, creating local clamping patterns that distort RNA structure and sharply decrease P_{RNA}. These mechanisms also explain changes in L_c, K and dL/dN of DNA and RNA duplexes.

The functions and applications of RNA often depend on its structures and structural transformations, which are influenced by the stiffness[49] and our results reveal that multivalent salts of high concentrations sharply reduce the stiffness of RNA duplex. We propose a physical mechanism which anions brought by high-concentration salts destroy the uniform cation clamping for the major groove of RNA duplexes, creating local cation-clamping that sharply distorts and softens the RNA duplex. While earlier studies generally emphasized cation effects[9,10,20,24,28,40], our findings highlight the significant influence of anions on nucleic acid structures and elasticities. This work greatly enriches the understanding of the structure and elasticity of RNA duplexes under extensive multivalent salt concentrations, which may be useful in rational control of RNA structures in material applications. Our findings extend beyond basic nucleic acid biophysics, potentially influencing fields such as RNA physics, drug development, and nanotechnology.

Methods

MT experiments

Nucleic acid constructs (DNA or RNA) were labeled with digoxigenin on one end and biotin on the other end (Fig. 1A). The glass slides were cleaned with piranha solution, sequentially coated with 1% APTES, 2% glutaraldehyde, and 1 mg/mL anti-digoxigenin, and then passivated with 2% BSA;

after glutaraldehyde coating, polystyrene beads were attached as reference points for drift compensation. Experiments were performed in an ultraclean room at 22 °C in 10 mM Tris-HCl buffer (pH 7.5). In the flow cell, the digoxigenin-labeled end was attached to the anti-digoxigenin-coated glass slide, while the biotin-labeled end was bound to a streptavidin-coated paramagnetic microbead (Dynabeads M-270). A pair of neodymium magnets applied force to stretch the construct, with force controlled by a linear motor (L-509, Physik Instrument); bead positions were tracked via diffraction patterns using a 100× oil immersion objective (Olympus) and a CCD camera (MER2-230-168U3M, Daheng Imaging). To minimize extension errors, molecules anchored near the center bottom of the beads were selected, and absolute extension was determined by measuring the glass height at zero force; the applied force (F) at each magnet height (d) was calibrated for each bead by fitting $F(d)$. Before force-extension measurements, all surfaces were passivated with mPEG-NHS ester in 100 mM sodium bicarbonate solution for 1 h to prevent nonspecific interactions.

DNA and RNA tethers for magnetic tweezers (Supplementary Fig. 3) were prepared from λ-DNA by PCR and, for RNA, additional T7 in vitro transcription. Torsion-free DNA was obtained by PCR with end-labeled primers (biotin and digoxigenin). Torsion-free RNA was assembled by annealing two T7-transcribed RNA strands with biotin- and digoxigenin-labeled DNA handles. For torsion-constrained constructs, multiply labeled biotin/digoxigenin handles were generated by PCR (or one-sided PCR for ssDNA handles) using biotin-11-dUTP or digoxigenin-11-dUTP, and then combined with PCR- or T7-derived long fragments to form torsion-constrained DNA or RNA tethers. Oligonucleotide sequences are provided in Supplementary Method 1.

All-atom MD simulations

We conducted all-atom MD simulations using the ssDNA sequence 5'-CGACTCTACGGCATCTGCGC-3' and the corresponding RNA sequence, where T was replaced with U, based on previous short DNA fragment experiments[24,40,50,51]. Initial structures were built using the nucleic acid builder (NAB) in AMBER[52]. Simulations were performed with Gromacs 4.6[53], using refined AMBER ff99bsc1 + χ_{OL3}[52,54] force fields and the TIP3P water model[55]. Additionally, respective salt ions (Ca^{2+}, Mg^{2+}, Sr^{2+}, SPD^{3+}, and Cl^{-}) at typical concentrations were added. The SPD^{3+} force field was parameterized based on the general AMBER force field[56], while the Ca^{2+}, Mg^{2+}, and Sr^{2+} cation models are taken from the work of Li et al. [57]. Meanwhile, we added Na^+ and Cl^- ions to ensure that our simulated systems were neutralized with the ion model from Joung and Cheatham[58]. Microscopic structural parameters were calculated with Curves+[59] (Supplementary Table 2), and system neutralization at desired salt concentrations was confirmed (Supplementary Fig. 6). Typical salt concentrations were chosen based on MT experimental results for both RNA and DNA. See more details of MD simulations in the Supplementary Method 2 and our previous works[43] for the simulations and the calculations of elastic parameters of RNA and DNA.

Statistics and reproducibility

For MT experiments, box plots show the data distribution across at least three independent molecules under each condition. For all-atom MD simulations, each condition was simulated for 600 ns, and box plots show the data distribution from four time intervals of the equilibrated trajectory.

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

The source data underlying the graphs are provided in Supplementary Data 1–4 in Excel format. The all-atom MD simulation files (including input files, topology/parameter files, and initial coordinates) are provided in Supplementary Data 5 in ZIP format.

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X.-H.Z. and Z.-J.T. conceived the idea. C.Z. and J.-H.Z. performed the experiments. H.L.D. performed the simulations. C.Z., H.L.D., L.D. and Y.Z. analyzed the data. All authors wrote the paper.

Competing interests

The authors declare no competing interests.

Additional information

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